



General

Introduction

Home Connect offers a RESTful API to control and monitor enabled home appliances. The following chapters provide details about the usage of this API. If it's your first time working with the Home Connect API, please see our [Quickstart](#) guide.

API client based on Swagger

Simulator

[API client - Simulated home appliances](#)

[Download API Specification](#)

Real API

[API client - Physical home appliances](#)

[Download API Specification](#)

Endpoints and Data Encoding



API of physical home appliances

api.home-connect.com or

api.home-connect.cn - This domain has to be used if you want to create an application for China. In this case please enable *Sync to China* in the [account settings](#) to copy all clients to China.

Data Encoding

The Home Connect API is based on HTTPS carrier with JSON data encoding.

All JSON transfer units are encoded with the following structure:

[> Expand source](#)

Best Practices

React on home appliance state

To get the full state of all paired home appliances of a user and keep your application updated with the latest state changes, you have to:

- Call [/api/homeappliances](#) to get the list of paired home appliances and evaluate if the home appliance is currently connected, i.e. online



settings information like the current *PowerState*

- If you want to show the currently active or selected program, request [/api/homeappliances/{hald}/programs/active](#) or [/api/homeappliances/{hald}/programs/selected](#), respectively.
- If required, request [/api/homeappliances/{hald}/images](#) on the fridge-freezer to get the list of stored images and then request [/api/homeappliances/{hald}/images/{imagekey}](#) for all 6 past images of interior and door camera. Thus, in total, these are 13 requests. If you also need the preview images, you have to do another 12 requests with the respective *previewimagekey*.
- Open the event monitoring channel for each home appliance on [/api/homeappliances/{hald}/events](#). The channel will stay open and you receive all changes of the home appliance right after they occurred. This includes any change which the user did locally on the home appliance as well as changes via the API.

Thus, in total you need 54 requests to get the full state of all 6 supported home appliances and be informed of each change.

Start program

In order to start a program, several pre-conditions have to be fulfilled:

- **Connection state:** The home appliance needs to be connected to the Home Connect server. The connection state can be retrieved by calling [/api/homeappliances](#) and reading the connected flag on the respective home appliance.
- **[BSH.Common.Status.RemoteControlActive](#):** The *RemoteControl* flag needs to be set to true. It must be set once on the appliance manually by the user.
- **[BSH.Common.Status.RemoteControlStartAllowed](#):** The *RemoteStart* flag needs to be set to true. It must be set on the appliance manually by the user.
- **[BSH.Common.Status.LocalControlActive](#):** The *LocalControl* flag needs to be set to false. It is true whenever the user operates the home appliance locally.
- **[BSH.Common.Status.OperationState](#):** The *OperationState* needs to be set to *BSH.Common.EnumType.OperationState.Ready*.



Pre-heat oven

If you want to pre-heat an oven, use the program [Cooking.Oven.Program.HeatingMode.PreHeating](#) with a duration longer than the oven requires to reach the target temperature. We recommend to use a minimum of 10 minutes. The oven will start a hot air program with the selected options.

Connect to the monitoring channel for this oven and wait for the [Cooking.Oven.Event.PreheatFinished](#) event. When you receive the event, notify the user that the oven is ready.

We recommend not to use the event *Cooking.Oven.Status.CurrentCavityTemperature* because the temperature might not provide the necessary accuracy.

Monitoring program progress

If you started a program on a home appliance, the home appliance will provide [status updates](#) on its event channel. Useful events are:

- [BSH.Common.Status.OperationState](#): Use this event to reflect the oven state in your application and identify manual user interaction. For instance, if the oven switches the `OperationState` to `_BSH.Common.EnumType.OperationState.Finished_` you can present a message to the user that the meal is ready. If the `OperationState` changes to `_BSH.Common.EnumType.OperationState.Pause_`, the user either pressed the Pause button or opened the door of the oven.
- Connection state: The home appliance is disconnected from the Home Connect server, e.g. due to a network error.



making a GET request, you have the ability to add the *Accept-Language* header in order to get all feature keys and enumeration values in the requested language.

Please note that API errors are not translated, e.g. if remote start disabled error is returned during program start.

Supported Languages

Language	Primary-Tag	Supported Language Tags
Bulgarian	bg	• bg-BG
Chinese (Simplified)	zh	• zh-CN • zh-HK • zh-TW
Croatian	hr	• hr-HR
Czech	cs	• cs-CZ
Danish	da	• da-DK
Dutch	nl	• nl-BE • nl-NL
English	en	• en-AU • en-CA • en-GB • en-IN • en-NZ



Finnish	fi	• fi-FI
French	fr	• fr-BE
		• fr-CA
		• fr-CH
		• fr-FR
		• fr-LU
German	de	• de-AT
		• de-CH
		• de-DE
		• de-LU
Greek	el	• el-GR
Hungarian	hu	• hu-HU
Italian	it	• it-CH
		• it-IT
Norwegian	nb	• nb-NO
Polish	pl	• pl-PL
Portuguese	pt	• pt-PT



Russian	ru	• ru-RU
Serbian	sr	• sr-SR
Slovak	sk	• sk-SK
Slovenian	sl	• sl-SI
Spanish	es	• es-CL
		• es-ES
		• es-PE
Swedish	sv	• sv-SE
Turkish	tr	• tr-TR
Ukrainian	uk	• uk-UA

Language tag follows the [HTTP/1.1 specification, section 3.10](#).

Example

Used Accept-Language header: *fr-FR*

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Rate Limiting

The Home Connect API uses the following rate limits. Please make sure not to reach the limits regularly, otherwise your client application may be blocked. At the moment two different rate limit types are supported.

Amount-based Rate Limits

Generally if too many requests are sent, further requests are rejected with the HTTP status code 429.

The following limits are currently configured:

- 10 event monitoring channels per client and Home Connect user account at any time (the request for further channels will be blocked)
- 10 requests per second on average (20 requests max. burst) with leaky bucket algorithm, otherwise HTTP 429 error

Example

No *Retry-After* header is set.

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Time-based Rate Limits

Generally if too many requests are sent, further requests are rejected with the HTTP status code 429. In this case, please check the *Retry-After* header. This states the number of seconds that you need to wait, before you try your request again.

The following limits are currently configured:

- 10 token refresh per minute and 100 token refresh per day



home appliance is allowed.

- 50 requests per client and Home Connect user account per minute (more requests will be blocked for one minute)
- 5 program starts per client and Home Connect user account per minute (more program starts will be blocked for one minute)
- 5 program stops per client and Home Connect user account per minute (more program stops will be blocked for one minute)
- 10 successive requests per client and Home Connect user account per 10 minutes which result in an error (more requests will be blocked for 10 minutes)

Example

Retry-After header is set to 52.

[Expand source](#)

API Errors

The Home Connect API returns appropriate HTTP status codes for every request.

Code	Short Description	Long Description
200	OK	The request was successful. Typically returned for succ requests.



204	No Content	PUT/DELETE requests with no payload.
400	Bad Request	Error occurred (e.g. validation error - value is out of range)
401	Unauthorized	No or invalid access token
403	Forbidden	Scope has not been granted or home appliance is not connected to HC account
404	Not Found	This resource is not available (e.g. no images on washing machine)
405	Method not allowed	The HTTP Method is not allowed for this resource
406	Not Acceptable	The resource identified by the request is only capable of generating response entities which have content characteristics not acceptable according to the accept headers sent in request.
408	Request Timeout	API Server failed to produce an answer or has no connection to backend service
409	Conflict	Command/Query cannot be executed for the home appliance, the error response contains the error details
415	Unsupported Media Type	The request's Content-Type is not supported
429	Too Many Requests	E.g. the number of requests for a specific endpoint exceeds quota of the client
500	Internal Server Error	E.g. in case of a server configuration error or any error in resource files
503	Service Unavailable	E.g. if a required backend service is not available

Errors may have a JSON payload with the following structure:

[Expand source](#)



401	401	Authentication is possible failed or not yet been pro
401	invalid_token	• The access token is malformed • The access token expires • The access token is • Token refers to an account.
404	404	The requested resource could not be found but may be available in the future.
405	405	HTTP method not allowed. Supported methods: {list supported methods}
404	SDK.Error.NotInSimulator	This endpoint / method is not implemented on the simulator. It may be valid for real home appliances.
429	429	The user has sent too many requests in a given amount of time.
403	insufficient_scope	• Insufficient scope for this resource • Not allowed to access this HomeAppliance • Action not supported for this HomeAppliance type • Insufficient permissions for this resource
409	SDK.Error.UnsupportedOperation	Operation not supported for this HomeAppliance type {HomeApplianceType}



Status		Message
415	415	The request's Content-Type is not supported. Expected: {list of supported types}
400	400	- The request content is malformed: {error description} - data.key does not match {key} - Request was malformed due to an invalid request path
409	SDK.Error.HomeAppliance.Connection.Initialization.Failed	- HomeAppliance is currently in a state that does not allow to respond to connect initialization request, it might be offline - HomeAppliance did not respond in time, it is currently offline
409	SDK.Error.UnsupportedProgram	Unknown program feature {featurekey}
404	SDK.Error.UnsupportedProgram	Program not supported
409	SDK.Error.UnsupportedOption	- Unknown option feature {featurekey} - Program option not supported
409	SDK.Error.UnsupportedStatus	Status not supported



409	SDK.Error.UnsupportedSetting	Setting not found for HomeAppliance
409	SDK.Error.UnsupportedCommand	Command not supported for HomeAppliance
409	SDK.Error.UnsupportedUnit	Unknown unit {unit} for featurekey {featurekey}
409	SDK.Error.UnsupportedKey	- The given event key is not white-listed - The given event key is unknown for this home appliance - The given event key is used due to disable change
409	SDK.Error.UnsupportedValue	- Settings value: {value} is not supported - Command value: {value} is not supported
409	BSH.Common.Error.RemoteControlNotActive	RemoteControl is not enabled
409	BSH.Common.Error.RemoteStartNotActive	RemoteStart is not enabled
409	SDK.Error.WrongOperationState	Request cannot be performed since OperationState is {list of expected operation states}; Request can be performed since array contains {current state}



		Request cannot be performed since OperationStatus is not {list of expected operation states}; Program cannot be aborted; Request cannot be performed since no program is set
409	BSH.Common.Error.LockedByLocalControl	Request cannot be performed temporarily! due to local user intervention
409	SDK.Error.Cooking.Oven.Status.FrontPanelOpen	FrontPanelOpen is open
409	SDK.Error.WrongDoorState	Request cannot be performed door is not closed
409	SDK.Error.Cooking.Oven.Status.MeatprobePlugged	Meatprobe is plugged
409	SDK.Error.BatteryLevelTooLow	- Request cannot be performed since the battery level is less than 10% - Request cannot be performed since the battery level is less than 20%
409	SDK.Error.ConsumerProducts.CleaningRobot.Status.Lifted	Request cannot be performed the robot is lifted
409	SDK.Error.ConsumerProducts.CleaningRobot.Status.DustBoxNotInserted	Request cannot be performed the dust box is not inserted
409	SDK.Error.ConsumerProducts.CleaningRobot.Status.AlreadyAtHome	Device is already connected to charging station.
409	SDK.Error.SelectedProgramNotSet	Request cannot be performed no program is selected



409	SDK.Error.ActiveProgramNotSet	Request cannot be performed because no active program is set
409	SDK.Error.WrongProgramState	Program cannot be aborted
409	SDK.Error.ProgramNotAvailable	Program currently not available
409	SDK.Error.ProgramNotWritable	Program can currently not be written
409	SDK.Error.InvalidOptionState	• {featurekey} valid only with NotAvailable • {featurekey} valid only with NotWritable
404	SDK.Error.NoProgramSelected	There is no program selected
404	SDK.Error.NoProgramActive	There is no program active
409	SDK.Error.MissingOptionValue	{featurekey} is missing



		value {value} is below minimum {min value}
		• Program option {featurekey} has value {value} exceeds {min value} • Program option {featurekey} has unexpected type {type}
409	SDK.Error.InvalidOptionValue	• {featurekey} validation failed with OutOfBounds • {featurekey} validation failed with NotWritable • {featurekey} validation failed with InvalidOption • Program option {featurekey} has value {value} does not match stepsize {stepsize}
409	SDK.Error.InvalidOptionCombination	If SteamAssistLevel is enabled THEN FastPreHeat must be switched off
409	SDK.Error.InvalidSettingState	{featurekey} validation failed with NotWritable
409	SDK.Error.InvalidSettingValue	Setting {featurekey} has unexpected type or value

Next Chapter

